

# Output Content (OC)

Score: 1/5 — Serious concern | Confidence: medium

**Benchmark:** SWITCHBOARD\_CONVERSATIONAL\_SPEECH | **Context:** NLP/AI Researchers — Digital Mental Health Dialogue Systems (US-Centric, Diverse Clinical Populations)

**Output Content definition:** Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Transcribers were lay workers producing orthographic transcriptions and rating naturalness/intelligibility [Q22, Q25]; no clinical professionals were involved and no inter-annotator agreement is reported for any classification task. The paper does not describe annotator demographics. For deployment evaluating CBT/substance-use dialogue intent for diverse clinical populations, this stands in stark contrast to AnnoMI (10 experienced MI practitioners as annotators [WEB-12, WEB-16]) and HOPE (therapist co-designed taxonomy [WEB-7]). The user's elicitation downplayed annotator-population mismatch (Q2/A2), but documented LLM failure modes — 'deceptive empathy' and cultural-cue misreading [WEB-15] and stereotypical AAVE production [WEB-14] — show that label correctness for culturally inflected counseling utterances is a substantive latent risk rather than a complementary concern. OC is MODERATE priority per elicitation, but evidence supports a low score.

| ID     | Evidence                                                                                                                                                                                                                                                                 |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q2     | "About 2500 conversations by 500 speakers from around the U.S. were collected automatically over T1 lines at Texas Instruments." (p.1)                                                                                                                                   |
| Q22    | "Transcribers were later asked to rate the naturalness of conversations they listened to, using a five point scale from "very natural" (1) to "forced or unnatural-sounding" (5)." (p.2)                                                                                 |
| Q25    | "After producing the text, the transcribers were asked to rate each conversation on a number of properties, such as the amount of background noise or static, difficulty in understanding the talkers, and degree to which the conversants stayed on one subject." (p.2) |
| WEB-7  | HOPE labels co-designed by therapists and counseling experts ( <a href="https://dl.acm.org/doi/10.1145/3488560.3498509">https://dl.acm.org/doi/10.1145/3488560.3498509</a> )                                                                                             |
| WEB-12 | AnnoMI annotated by 10 experienced MI practitioners; explicitly examines IAA variation ( <a href="https://www.mdpi.com/1999-5903/15/3/110">https://www.mdpi.com/1999-5903/15/3/110</a> )                                                                                 |
| WEB-14 | LLMs overproduce stereotypical AAVE features in clinical contexts ( <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC12037967/">https://pmc.ncbi.nlm.nih.gov/articles/PMC12037967/</a> )                                                                                |
| WEB-15 | LLM CBT counselors produce 'deceptive empathy' and misread cultural cues ( <a href="https://arxiv.org/pdf/2409.02244">https://arxiv.org/pdf/2409.02244</a> )                                                                                                             |
| WEB-16 | AnnoMI publicly available with expert annotation protocol ( <a href="https://github.com/uccollab/AnnoMI">https://github.com/uccollab/AnnoMI</a> )                                                                                                                        |

## Strengths

- Naturalness, background noise, intelligibility, and topical-coherence ratings were systematically collected [Q22, Q25] with a documented protocol (mean 1.48 [Q23]) — methodologically rigorous for the original collection task.

| ID  | Evidence                                                                                                                                                                                                                                                                 |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q22 | "Transcribers were later asked to rate the naturalness of conversations they listened to, using a five point scale from "very natural" (1) to "forced or unnatural-sounding" (5)." (p.2)                                                                                 |
| Q23 | "The average rating of 1.48 suggests that the collection protocol was successful in this respect." (p.2)                                                                                                                                                                 |
| Q25 | "After producing the text, the transcribers were asked to rate each conversation on a number of properties, such as the amount of background noise or static, difficulty in understanding the talkers, and degree to which the conversants stayed on one subject." (p.2) |

## Checklist

**Checklist item definitions:**

| ID   | Assessment Question                                               |
|------|-------------------------------------------------------------------|
| OC-1 | Do ground-truth labels reflect regional stakeholder perspectives? |
| OC-2 | Assess potential annotator-regional population disagreement       |
| OC-3 | Review annotator demographics from documentation                  |
| OC-4 | Consider label re-annotation by regional annotators               |
| OC-5 | Review aggregation methods for minority perspective erasure       |
| OC-6 | Document label issues harming convergent/external validity        |

### Checklist responses:

**OC-1:** INSUFFICIENT DOCUMENTATION — the paper documents only orthographic transcription and naturalness ratings [Q22, Q25], not dialogue-act ground truth; therefore no labels exist that could be assessed for regional-stakeholder alignment within this corpus.

**OC-2:** Disagreement between original (lay, non-clinical) annotators and the target clinical/community population is highly likely for any culturally inflected counseling utterance [WEB-14, WEB-15]; clinical NLP comparators use clinical experts [WEB-7, WEB-12].

**OC-3:** INSUFFICIENT DOCUMENTATION — the paper does not report annotator demographics, training, or selection criteria beyond their role as transcribers [Q22, Q25].

**OC-4:** Re-annotation by representative regional clinical annotators is required for any counseling deployment use; AnnoMI's 10-MI-practitioner model [WEB-12, WEB-16] and HOPE's therapist co-design [WEB-7] are the methodological precedents.

**OC-5:** INSUFFICIENT DOCUMENTATION — no aggregation method for classification labels is reported because no classification labels are produced in this paper.

**OC-6:** Convergent and external validity for the counseling deployment cannot be supported by the existing annotation; lay-transcriber output does not generalize to clinical-intent labels for diverse communities [WEB-7, WEB-12, WEB-14, WEB-15].

| ID     | Evidence                                                                                                                                                                                                                                                                 |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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### Information Gaps

- Annotator demographics, training, IAA, and selection criteria are entirely undocumented in the paper.

### Requires Expert Verification

- Whether SWBD-DAMSL transferred annotations (developed externally in 1997) inherit lay-transcription bias when reused for clinical purposes.

### Remediation

**Gap 1: Lay transcribers, no clinical involvement, no IAA reported, no annotator demographic documentation [Q22, Q25].**

**Recommendation:** For any deployment-facing evaluation subset, re-annotate with clinical experts following AnnoMI's protocol ( $\geq 10$  trained MI/CBT practitioners) [WEB-12, WEB-16] and report stratified IAA across community-representative annotators.

| ID     | Evidence                                                                                                                                                                                                                                                                 |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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